

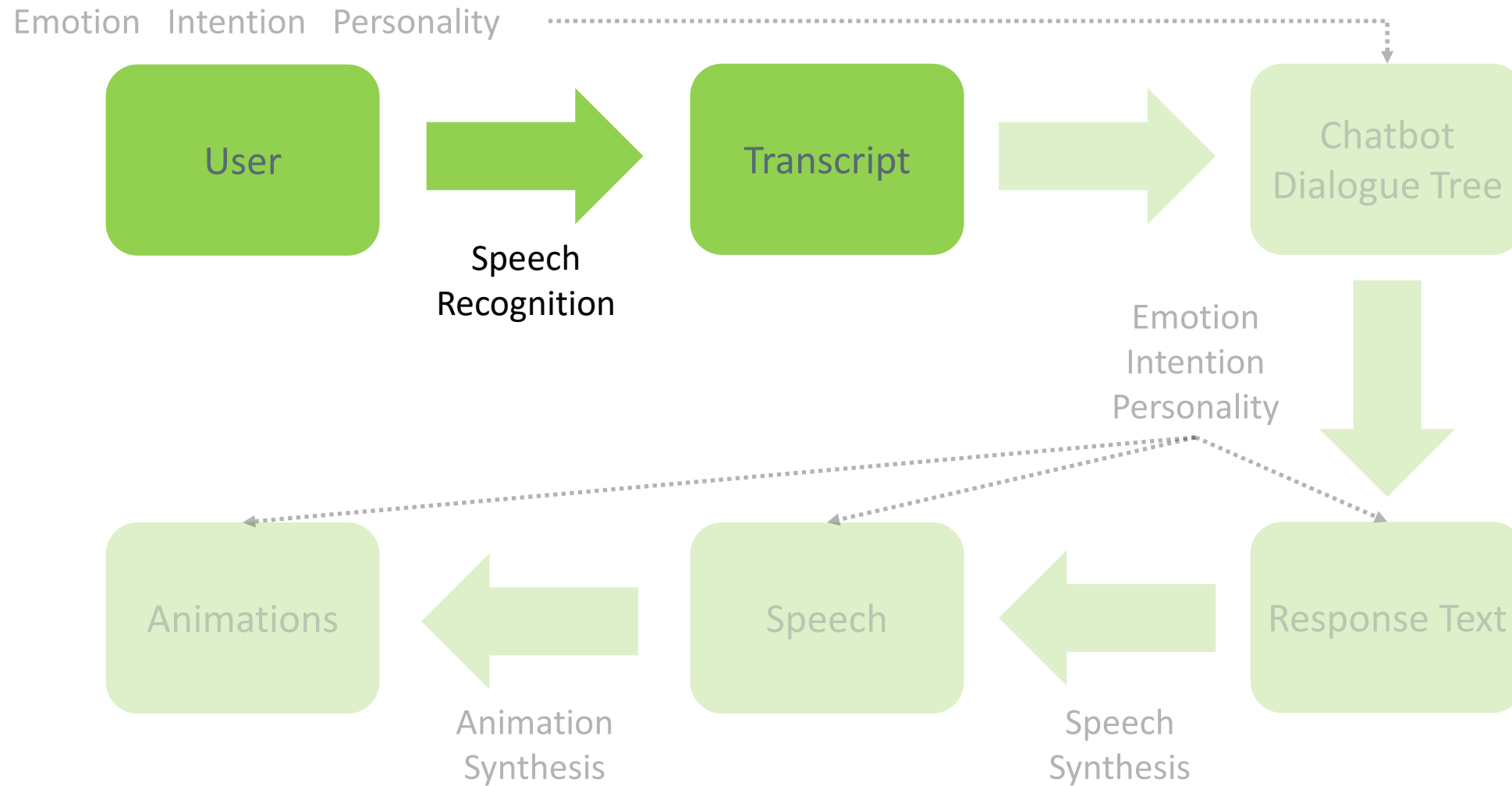


Artificial Intelligence for Digital Characters

Speech Recognition

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Conversational Digital Character



Phonetics

Feature Extraction

Architectures

Word Error Rate

Phones

Prosody

Pitch

Frequency Domain

Phonetic Alphabet

Phones: speech sounds, each represented with symbols

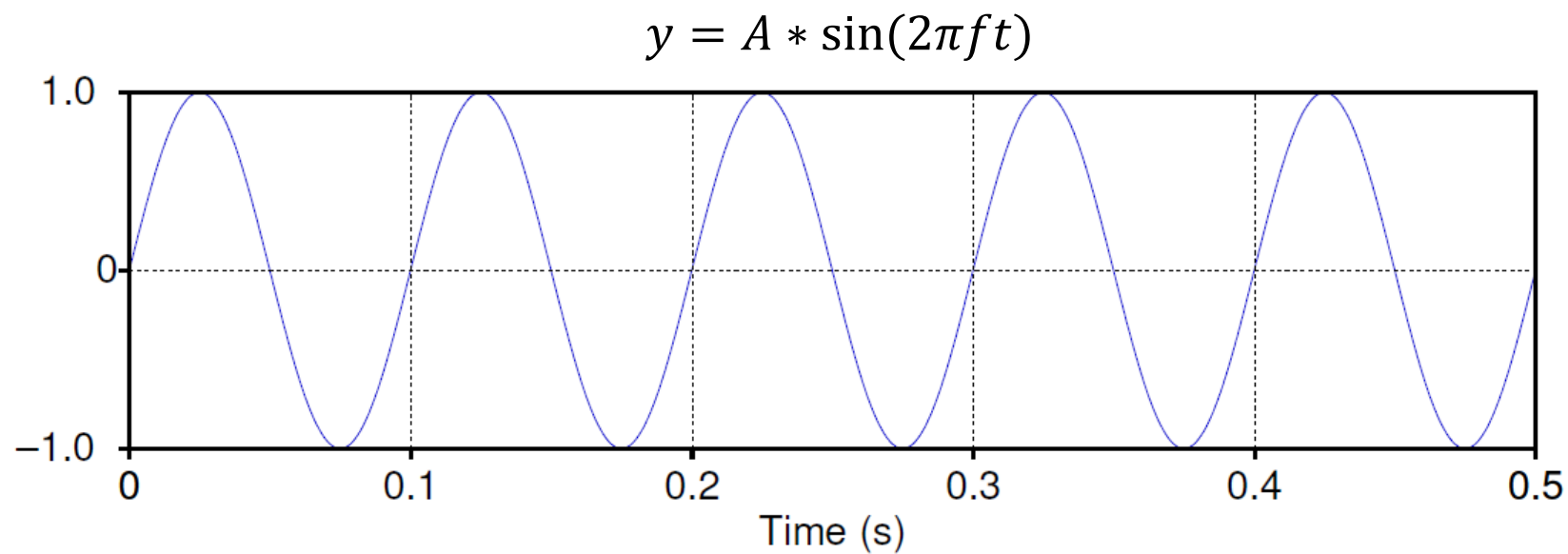
String of phones: Pronunciation of a word

ARPAbet | IPA

Alphabets for transcribing American English phonetic sounds

ARPAbet Symbol	IPA Symbol	Word	ARPAbet Transcription	ARPAbet Symbol	IPA Symbol	Word	ARPAbet Transcription
[p]	[p]	<u>p</u> arsley	[p aa r s l iy]	[iy]	[i]	lily	[l ih l iy]
[t]	[t]	<u>t</u> ea	[t iy]	[ih]	[ɪ]	lily	[l ih l iy]
[k]	[k]	<u>c</u> ook	[k uh k]	[ey]	[eɪ]	da <u>s</u> ey	[d ey z iy]
[b]	[b]	<u>b</u> ay	[b ey]	[eh]	[ɛ]	pe <u>n</u>	[p eh n]
[d]	[d]	<u>d</u> ill	[d ih l]	[ae]	[æ]	as <u>t</u> er	[ae s t axr]
[g]	[g]	<u>g</u> arlic	[g aa r l ix k]	[aa]	[ɑ]	pop <u>p</u> y	[p aa p iy]
[m]	[m]	<u>m</u> int	[m ih n t]	[ao]	[ɔ]	or <u>ch</u> id	[ao r k ix d]
[n]	[n]	<u>n</u> utmeg	[n ah t m eh g]	[uh]	[ʊ]	wo <u>o</u> d	[w uh d]
[ng]	[ŋ]	ba <u>k</u> ing	[b ey k ix ng]	[ow]	[oʊ]	lot <u>u</u> s	[l ow dx ax s]
[f]	[f]	<u>f</u> lour	[f l aw axr]	[uw]	[u]	tul <u>i</u> p	[t uw l ix p]
[v]	[v]	clo <u>v</u> e	[k l ow v]	[ah]	[ʌ]	bu <u>t</u> ter	[b ah dx axr]
[th]	[θ]	<u>t</u> hick	[th ih k]	[er]	[ɜː]	bi <u>r</u> d	[b er d]
[dh]	[ð]	<u>t</u> hose	[dh ow z]	[ay]	[aɪ]	i <u>r</u> is	[ay r ix s]
[s]	[s]	<u>s</u> oup	[s uw p]	[aw]	[aʊ]	flow <u>e</u> r	[f l aw axr]
[z]	[z]	egg <u>s</u>	[eh g z]	[oy]	[oɪ]	so <u>i</u> l	[s oy l]
[sh]	[ʃ]	squash <u>e</u>	[s k w aa sh]	[ax]	[ə]	pit <u>a</u>	[p iy t ax]
[zh]	[ʒ]	ambrosia	[ae m b r ow zh ax]				
[ch]	[tʃ]	<u>ch</u> erry	[ch eh r iy]				
[jh]	[dʒ]	<u>j</u> ar	[jh aa r]				
[l]	[l]	<u>l</u> icorice	[l ih k axr ix sh]				
[w]	[w]	ki <u>w</u> i	[k iy w iy]				
[r]	[r]	<u>r</u> ice	[r ay s]				
[y]	[j]	<u>y</u> ellow	[y eh l ow]				
[h]	[h]	<u>h</u> oney	[h ah n iy]				

Acoustic Signals: Sine & Cosine

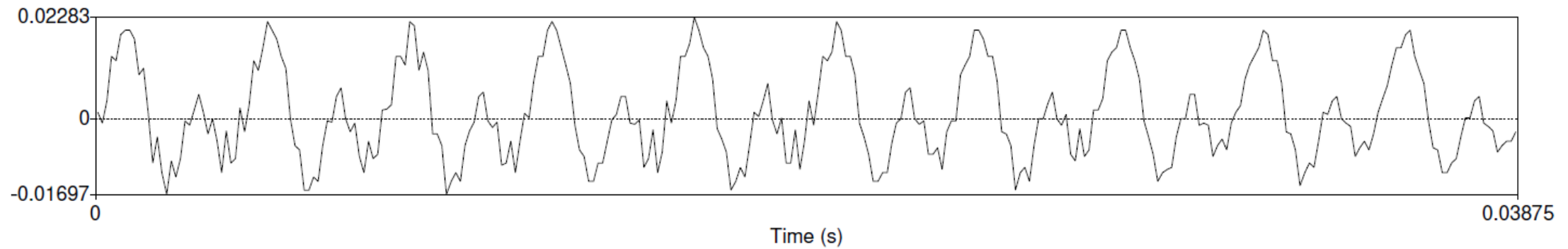


frequency $f = 10$ Hz (cycles per second)

amplitude $A = 1$ (max value on y axis)

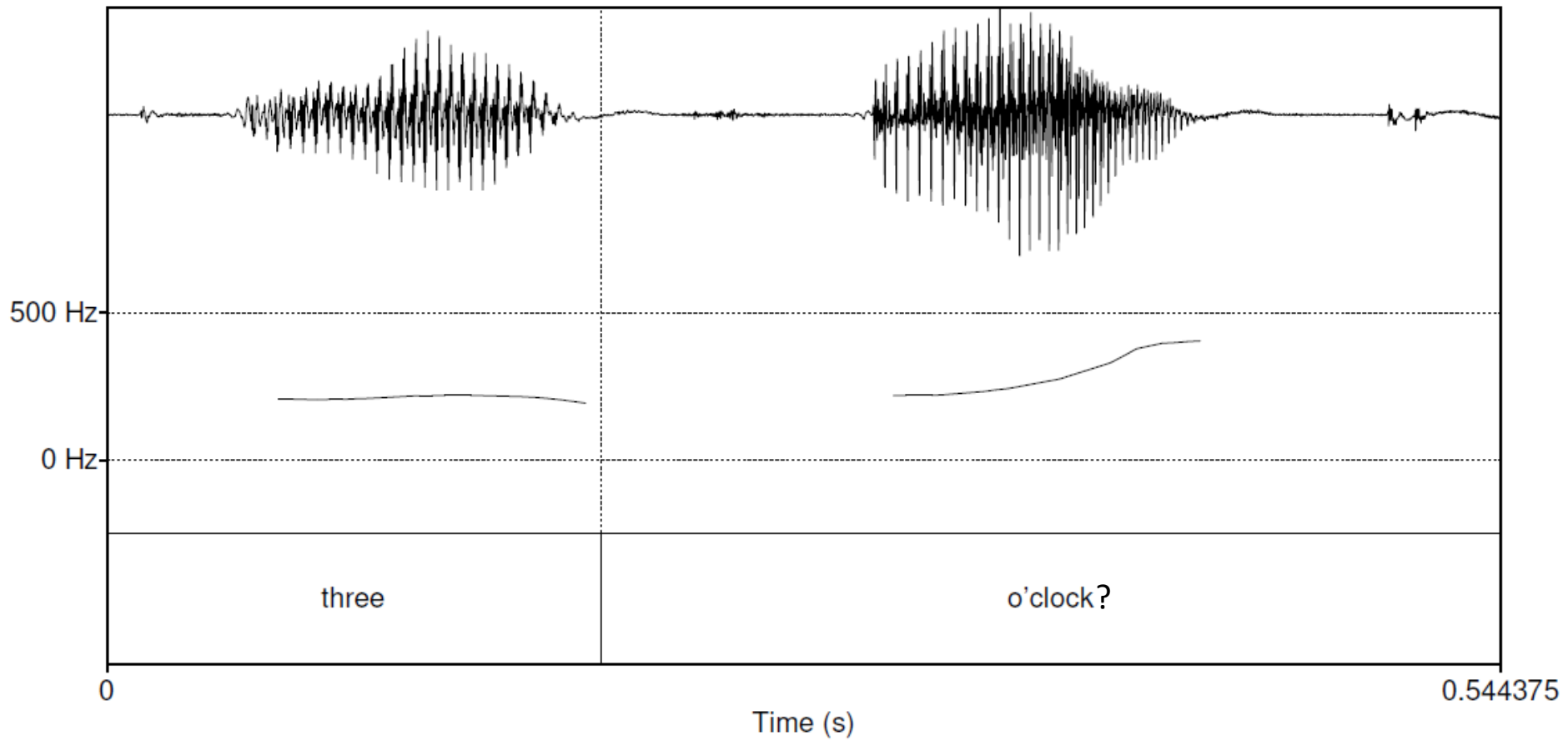
period $T = 1 / f = 0.1$ s (time for one cycle)

Waveform of the phoneme [iy] (she just had a baby^y)



Frequency: 10 repetitions / .03875 seconds = 258 Hz

Pitch Track

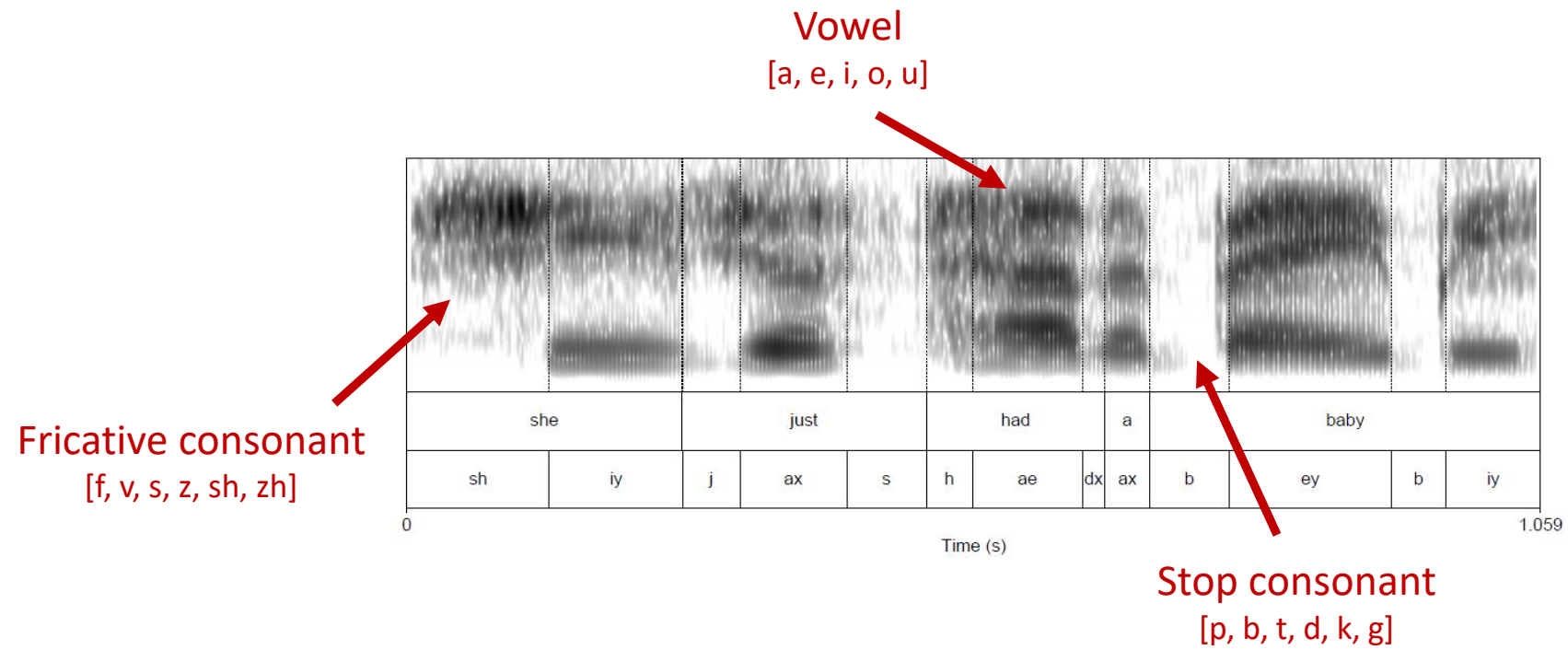


Pitch is not Frequency

- Pitch is the mental sensation or perceptual correlate of F0 (F0 = lowest frequency component of a complex waveform)
- Relationship between pitch and F0 is not linear
 - human pitch perception is most accurate between 100Hz and 1000Hz.
 - Linear in this range
 - Logarithmic above 1000Hz
- Mel scale is one model of this F0-pitch mapping
 - A mel is a unit of pitch defined so that pairs of sounds which are perceptually equidistant in pitch are separated by an equal number of mels
 - Frequency in mels = $1127 \ln(1 + f/700)$
 - Mel scale important for speech recognition

- Every complex wave can be represented as a sum of many sine waves of different frequencies (F_0 = lowest frequency component)
- Spectrum: Frequency components and amplitudes of a signal
- Phones have characteristic spectral signatures (peaks)
- Spectral representations important for speech recognition
- Spectrogram (frequencies over time)
 - Time (x-axis)
 - Frequencies (y-axis)
 - Color coding (amplitude)

Frequency Domain: Spectrogram

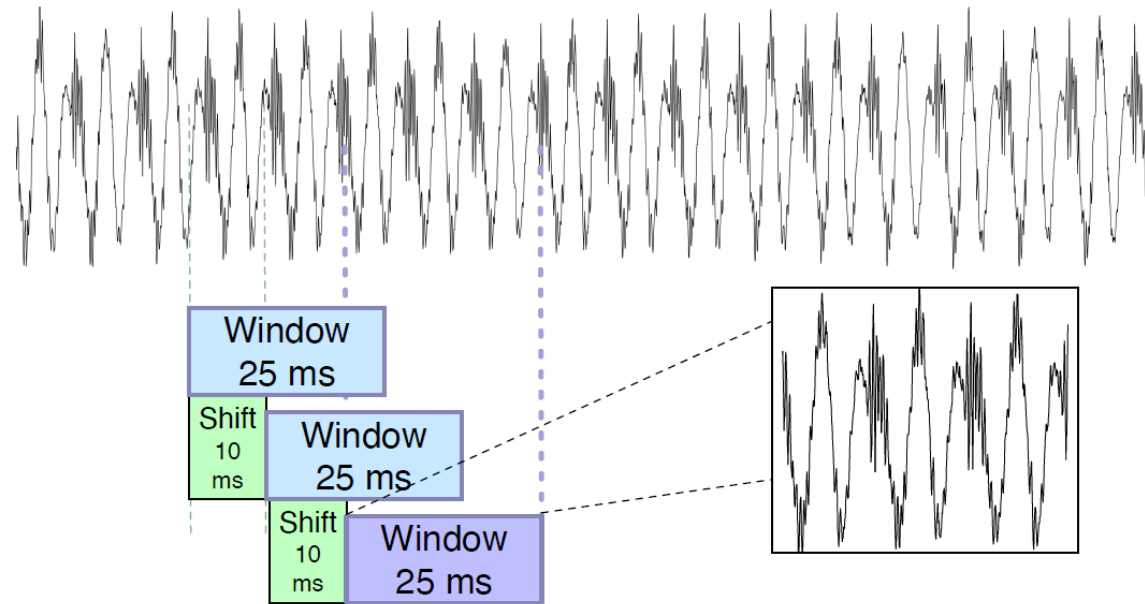


- Software for analysis of speech
 - Spectral analysis
 - Pitch tracking
- <https://www.fon.hum.uva.nl/praat/>
(Mac, Windows, Linux)

Feature Extraction

Phonetics
Architectures
Word Error Rate

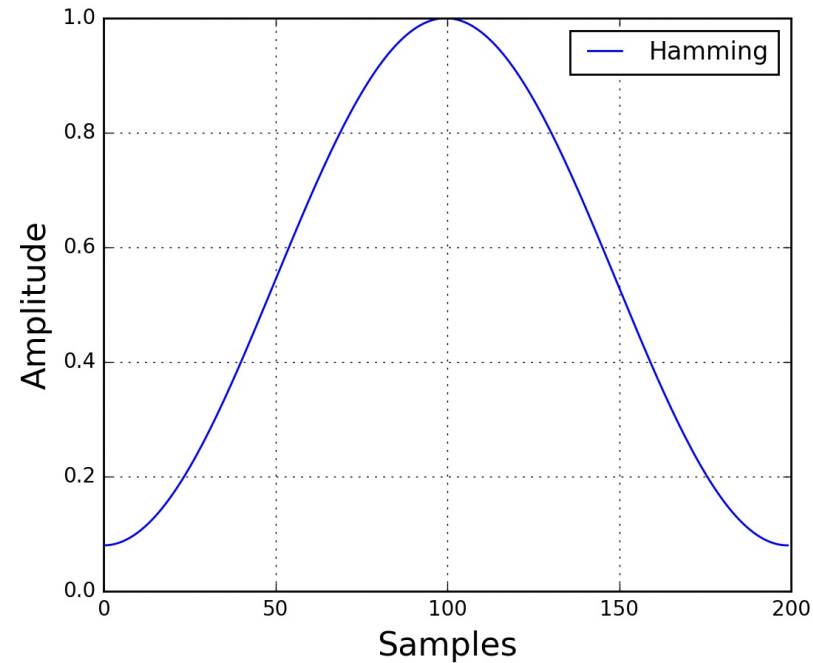
Windowing
Discrete Fourier Transform
Mel Filter Bank
Augmentation



- Frequencies in a signal change over time
- Frequencies in a signal stationary over short period of time
- Frame: Speech in a window
- Parameters:
 - Window size: typically between 20ms to 40ms
 - Stride of window: typically 50% (+/- 10%)
 - Shape of window

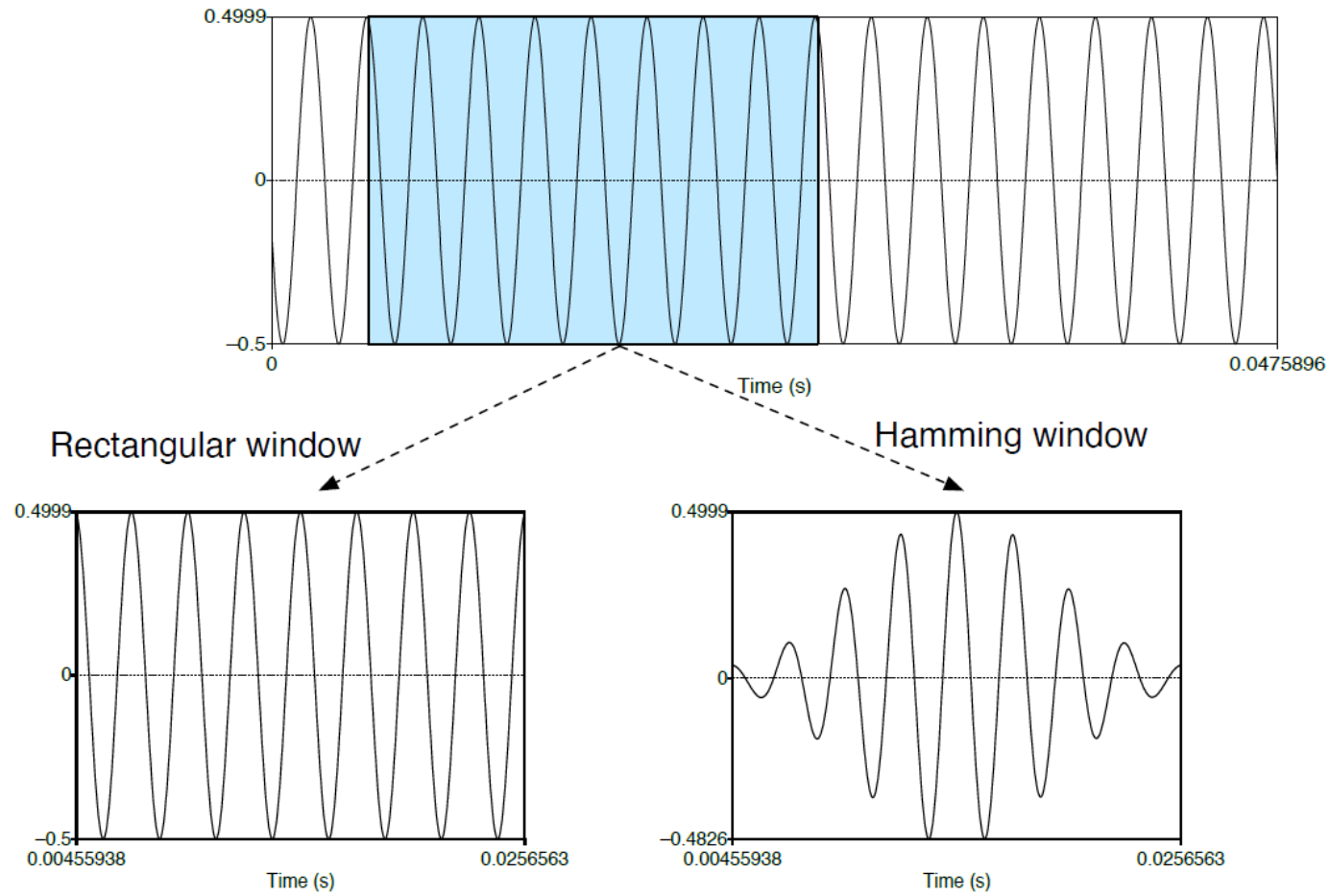
Shape of Window: Hamming Window

$$w[n] = 0.54 - 0.46 \cos \frac{2\pi n}{N-1} \quad 0 \leq n \leq N-1$$

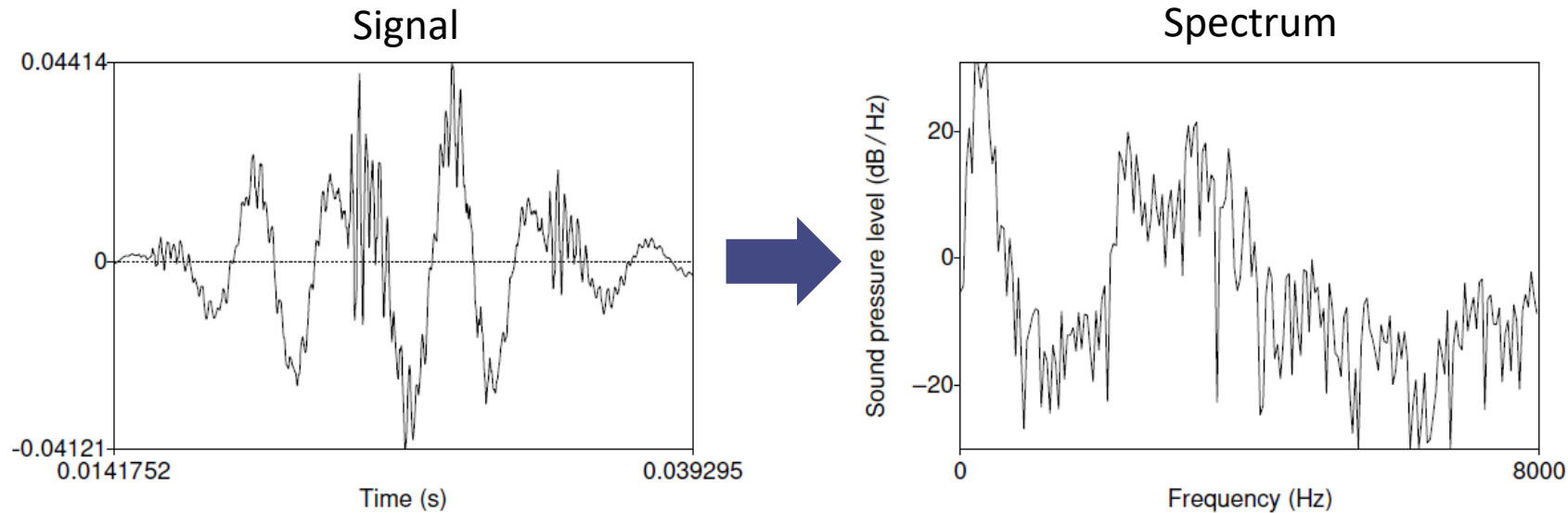


Applied to each window → Minimizing discontinuities and spectral leakage

Shape of Window

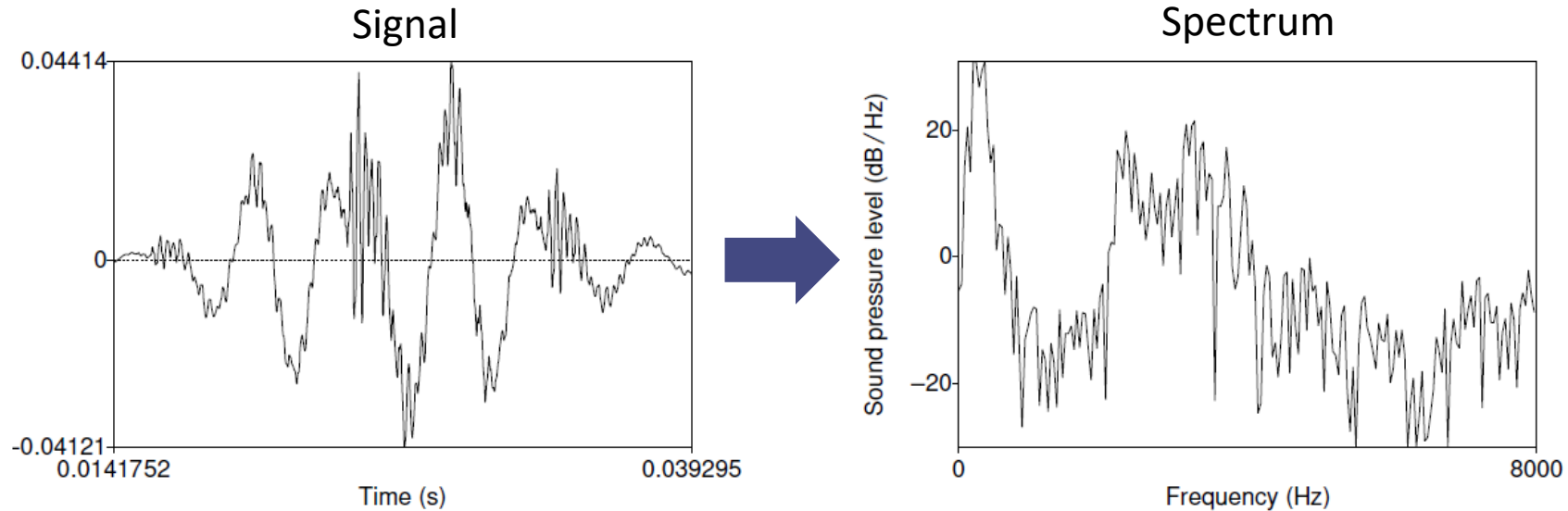


Discrete Fourier Transform



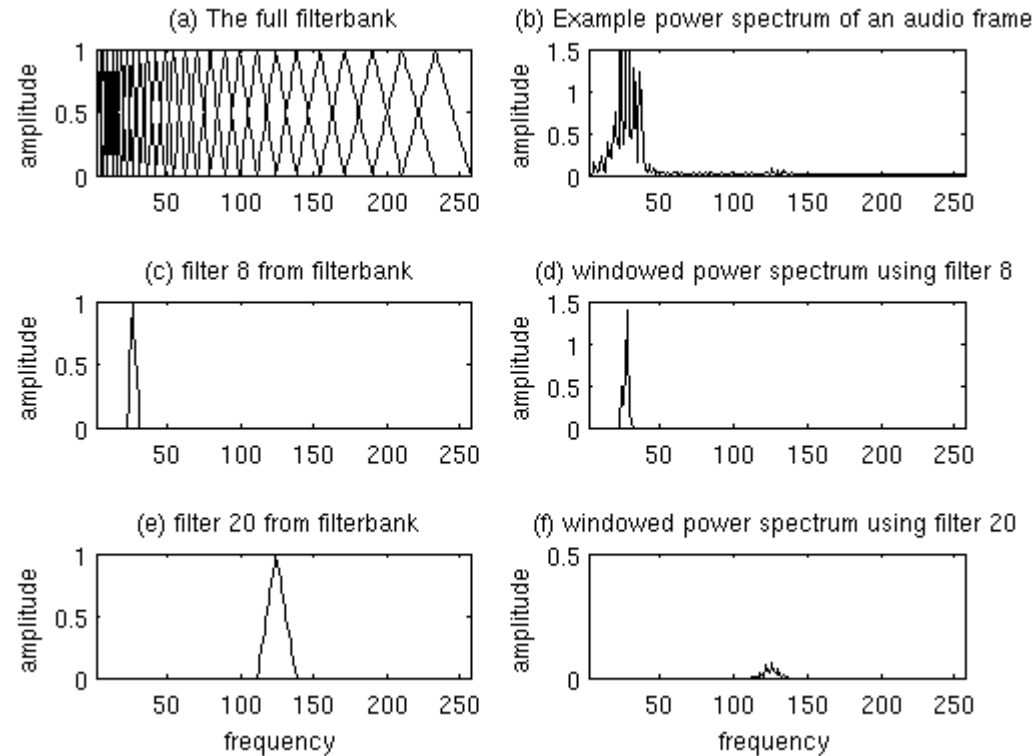
- Extract spectral information for each window
- Discrete Fourier transform: $X(k) = \sum_{n=0}^{N-1} x(n) * e^{-j\frac{2\pi}{N}kn}$
 - $x(n)$: Signal with N samples
 - k : Index of the frequency bin, which ranges from 0 to $N-1$
 - N : Typically 256 or 512
 - $X(k)$: Complex number representing magnitude and phase of frequency bin k
- Fast Fourier transform (FFT)
 - Very efficient calculation of DFT
 - Works only for values of N that are powers of 2

Discrete Fourier Transform



- $f_k = k * \frac{f_{sample}}{N}$
 - f_k : Frequency corresponding to $X(k)$
 - f_{sample} : Sampling rate of the signal
- Power spectrum: $P(k) = \frac{|X(k)|^2}{N}$

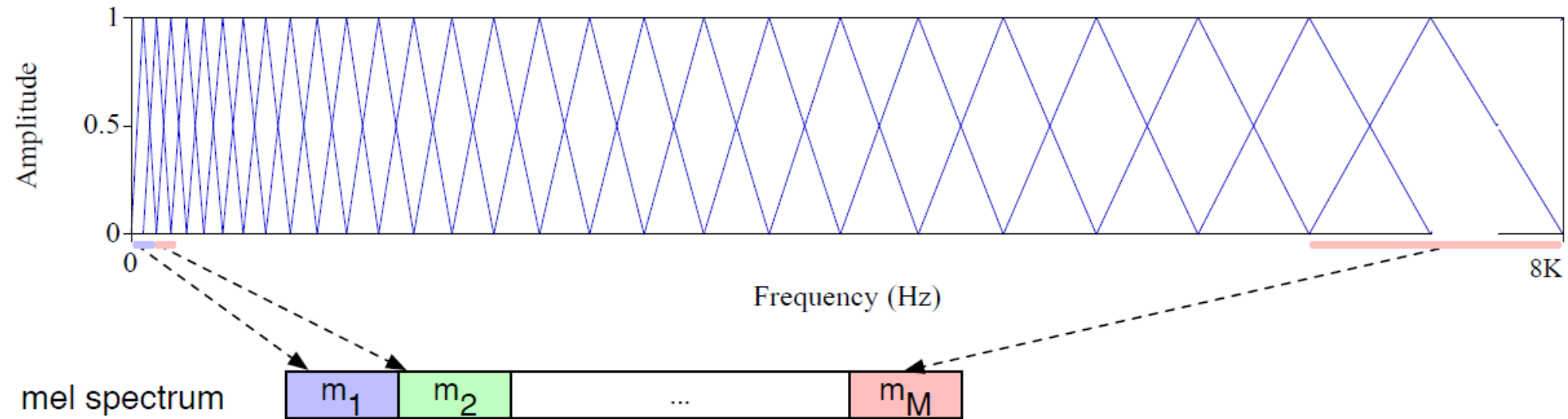
Mel Filter Bank



For each frame...

- we have values $P(k)$, $k: 1 \dots K$
- apply triangular filters on a Mel-scale to extract frequency bands
- take the log of each of the mel spectrum values

Mel Filter Bank



Mel Filter bank

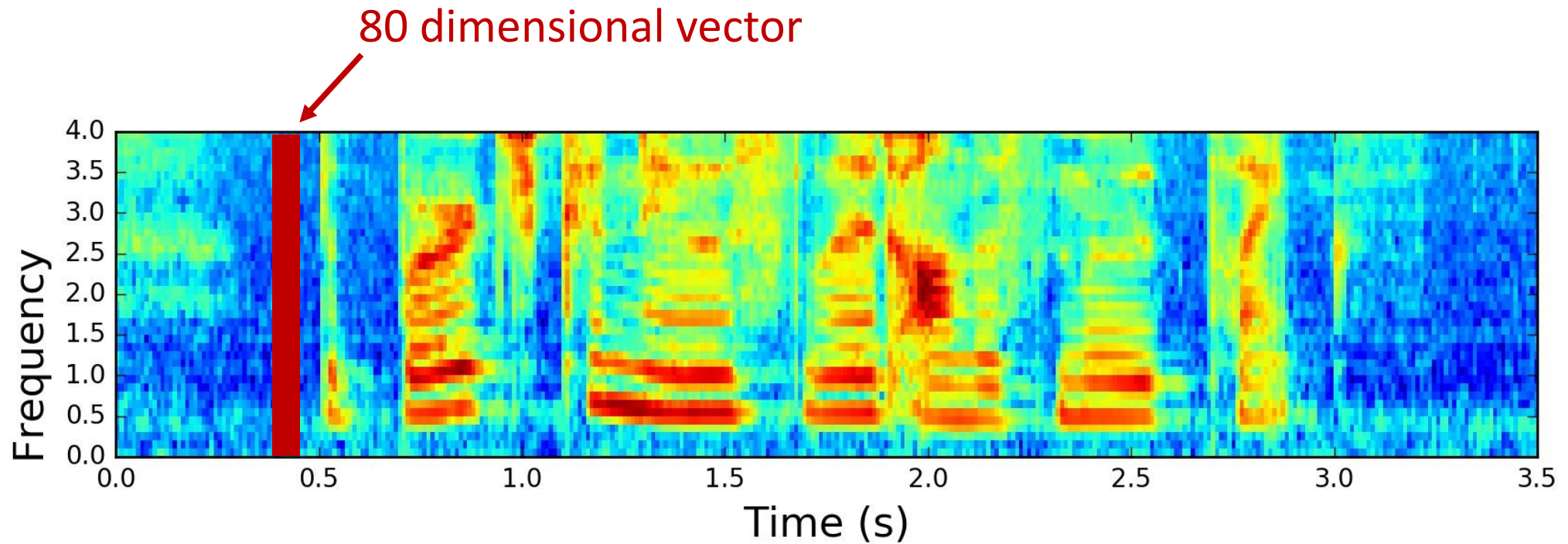
- Roughly uniformly spaced before 1 kHz
- Logarithmic scale after 1 kHz

How to Construct Mel Filter Bank

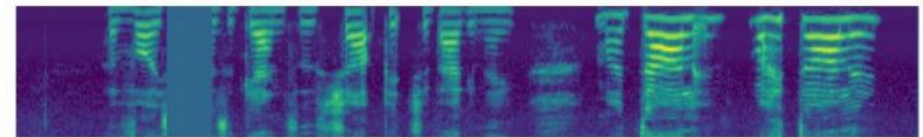
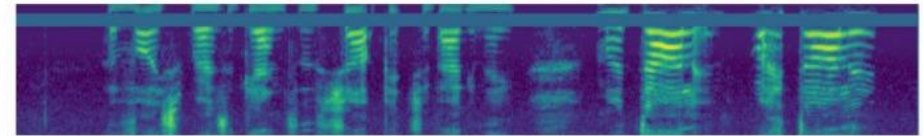
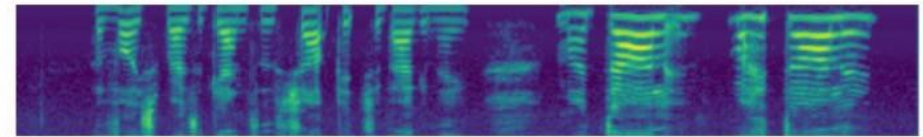
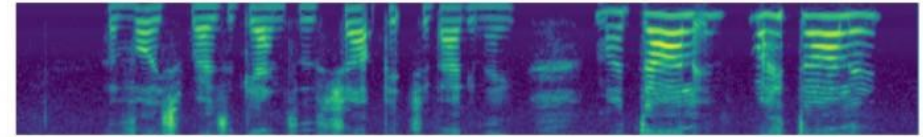
1. Convert lower and upper frequency to Mels Mels = $1127 \ln(1 + f/700)$
2. Space points linearly between lower and upper frequency
(e.g., for 80 filterbanks, add 80 additional points)
3. Convert points back to Hertz
4. Round frequencies to nearest frequency bin
5. Create filterbanks
First filterbank starts at first point, peaks at second point and returns to zero at third point etc.

$$H_m(k) = \begin{cases} 0 & k < f(m-1) \\ \frac{k - f(m-1)}{f(m) - f(m-1)} & f(m-1) \leq k \leq f(m) \\ \frac{f(m+1) - k}{f(m+1) - f(m)} & f(m) \leq k \leq f(m+1) \\ 0 & k > f(m+1) \end{cases}$$

Spectrogram



- Time warping
 - Distorting the time axis
- Frequency masking
 - Masking certain frequency bands
- Time masking
 - Masking certain time frames



Phonetics
Feature Extraction

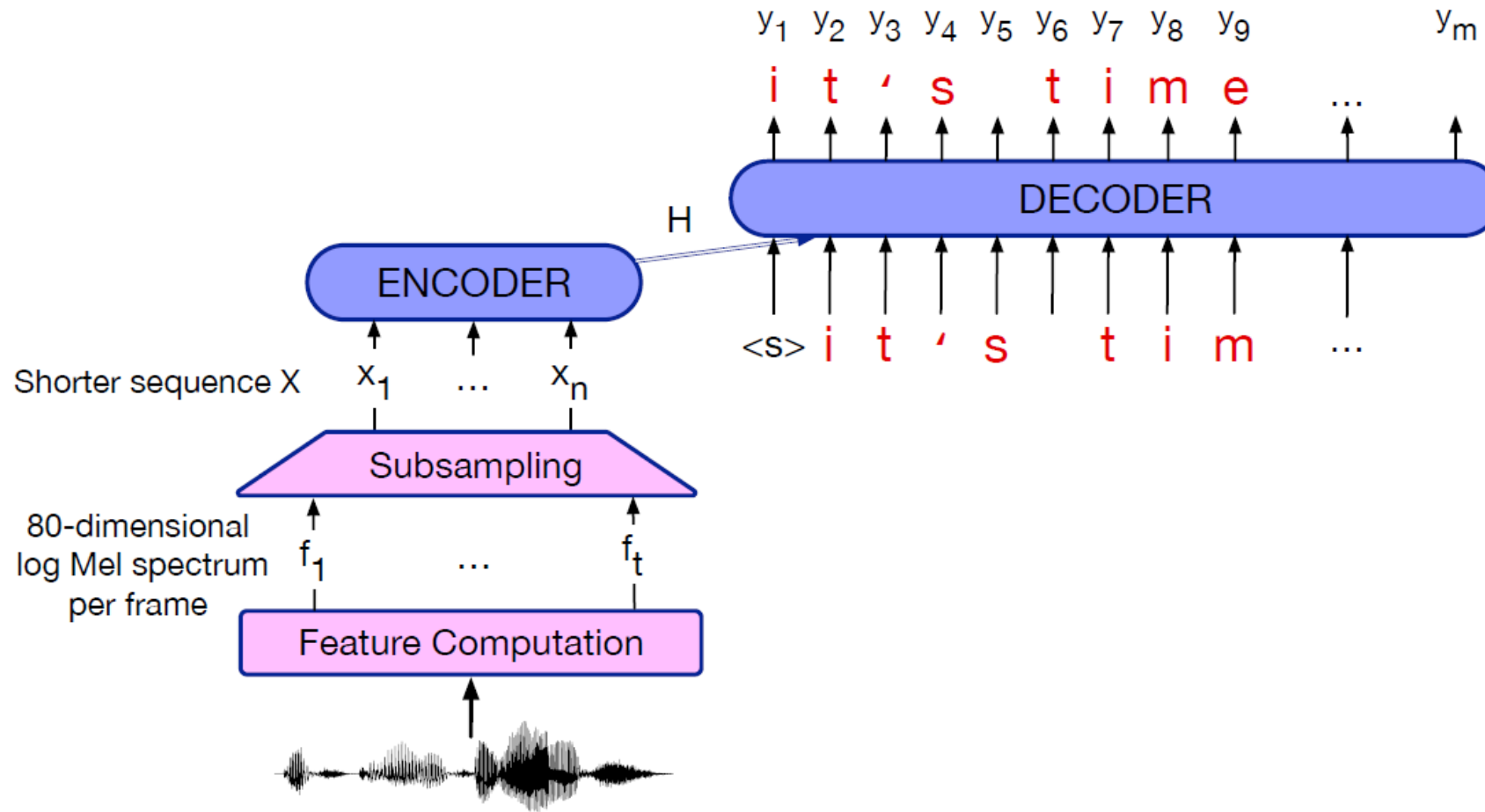
Architectures

Word Error Rate

Encoder-Decoder
Whisper
CTC
RNN-Transducer

Encoder-Decoder Speech Recognizer

$y_i \in \{a, b, c, \dots, z, 0, \dots, 9, < \text{space} >, < \text{comma} >, < \text{period} >, < \text{apostrophe} >, < \text{unk} >\}$



- Size input >> size output
- Compression to shorten acoustic feature sequence
- Algorithms:
 - Low frame rate: $f_i = \text{concat}(f_i, f_{i-1}, f_{i-2})$, delete f_{i-1} and f_{i-2}
 - Convolutional net (downsamples with max pooling)
 - Pyramidal RNNs (each successive layer has half the number of time steps)

Whisper (Model Architecture)

Multitask training data (680k hours)

English transcription

“Ask not what your country can do for ...”
Ask not what your country can do for ...

Any-to-English speech translation

“El rápido zorro marrón salta sobre ...”
The quick brown fox jumps over ...

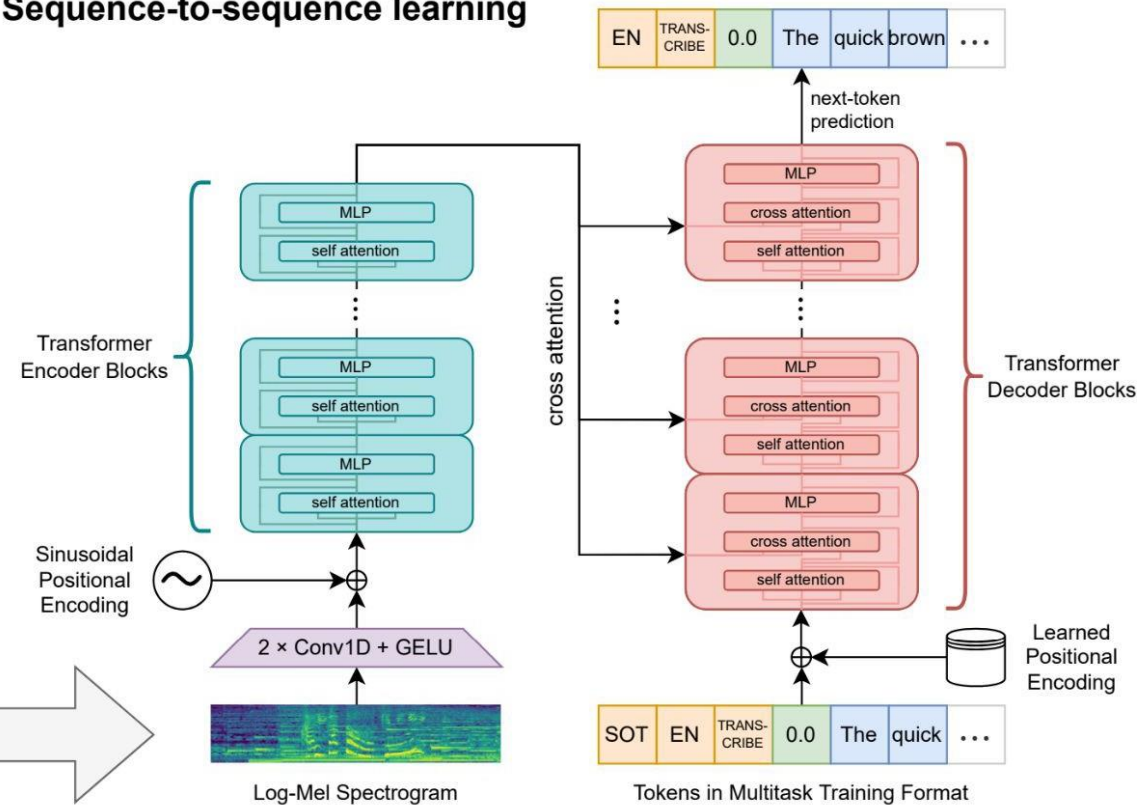
Non-English transcription

“언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...”
언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...

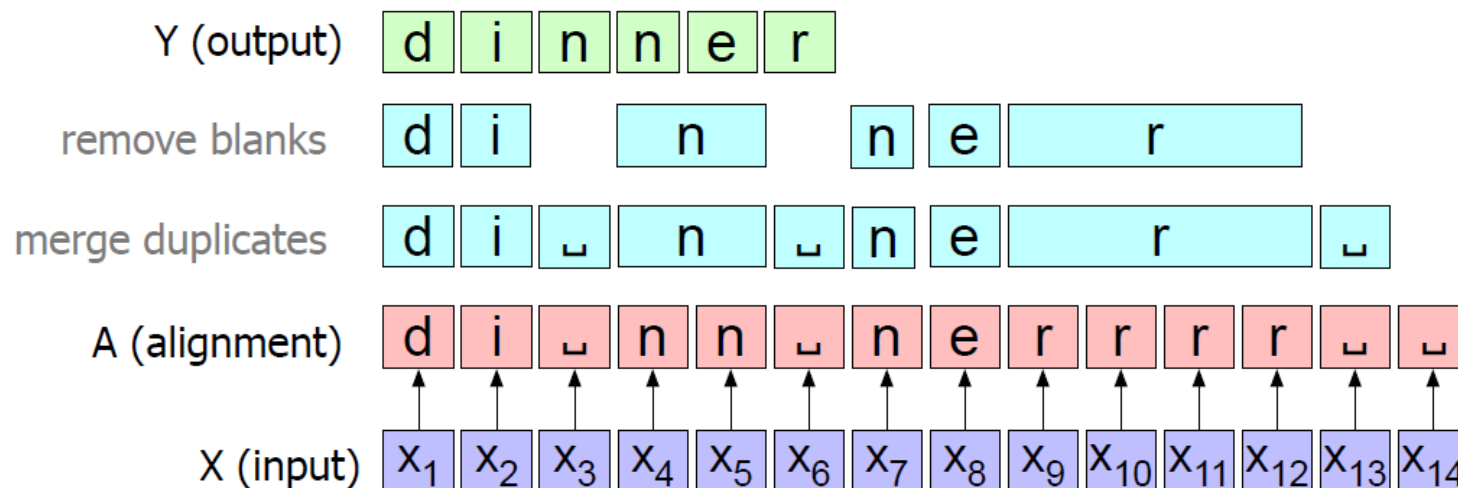
No speech

(background music playing)
∅

Sequence-to-sequence learning



Connectionist Temporal Classification (CTC)



- Output a single character for every frame x_i (alignment)
- Special symbol for a blank (e.g., silence)
- Collapsing function collapses consecutive duplicate letters
- Collapsing function is many-to-one: Different alignments map to same output string
- Assumption: Output at time t is independent of the output labels at any other time

$$P_{CTC}(A|X) = \prod_{t=1}^T p(a_t|X)$$

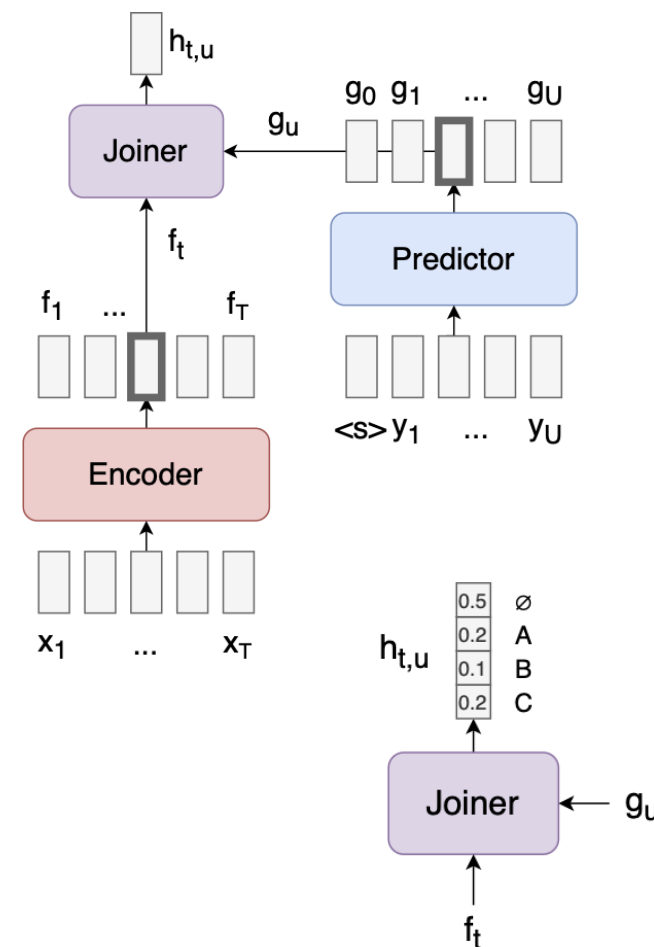
$$a_t = \operatorname{argmax}_{c \in \mathcal{C}} p_t(c|X) \quad A = \{a_1, \dots, a_T\}$$

- Problem: Most likely alignment A may not correspond to the most likely final collapsed output string Y
- Pick output string Y that has highest sum over the probability of all its possible alignments
- Approximation of sum of probabilities
 - Training: Dynamic programming

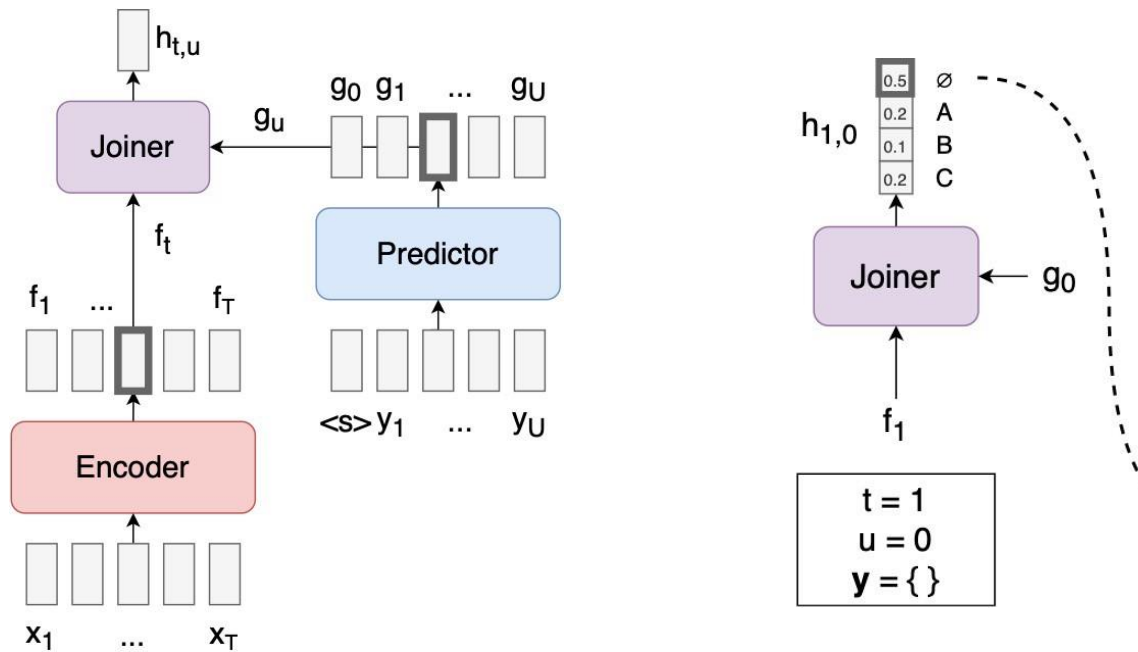
$$\begin{aligned} P_{CTC}(Y|X) &= \sum_{A \in B^{-1}(Y)} P(A|X) \\ &= \sum_{A \in B^{-1}(Y)} \prod_{t=1}^T p(a_t|h_t) \\ \hat{Y} &= \operatorname{argmax}_Y P_{CTC}(Y|X) \end{aligned}$$

Three components

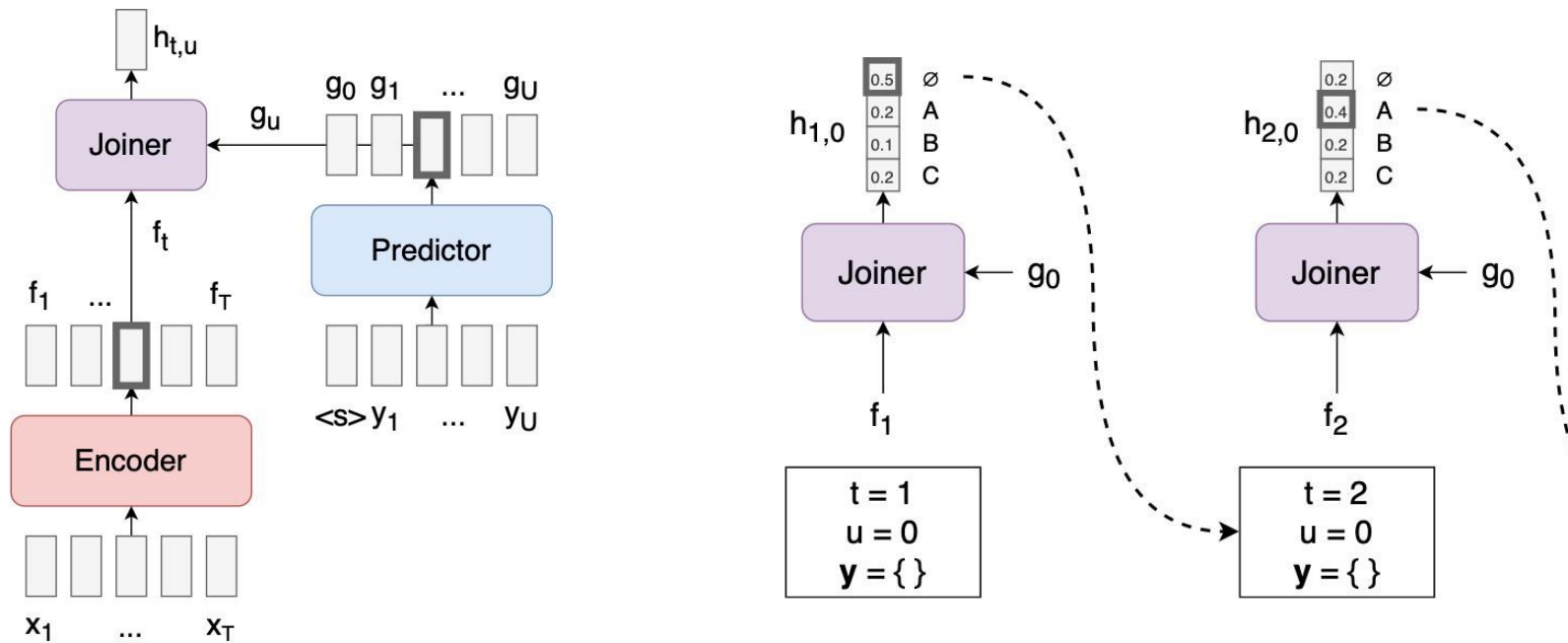
- CTC acoustic model
- Predictor (language model)
 - Conditions on sequence output so far
 - Has access to y and not x (unlike attention model)
 - Pre-train on text-only data
- Joiner
 - feedforward network



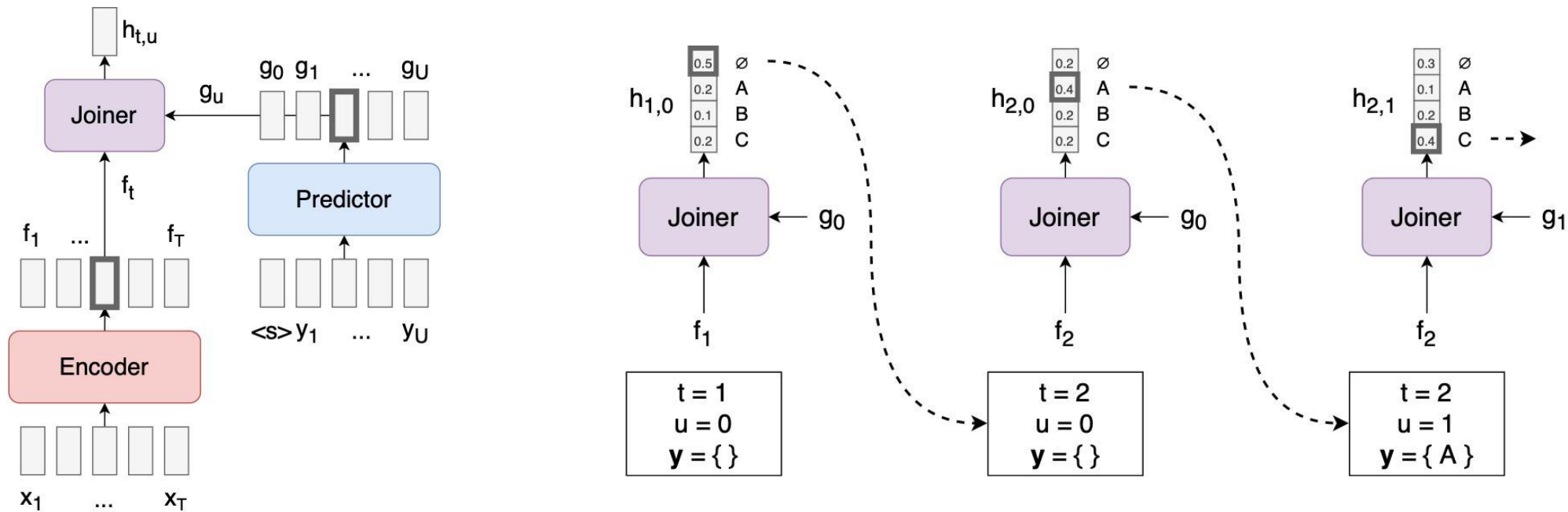
- Works for online decoding (streaming)
- Increment u when non-blank token is output
- increment t when blank token is output



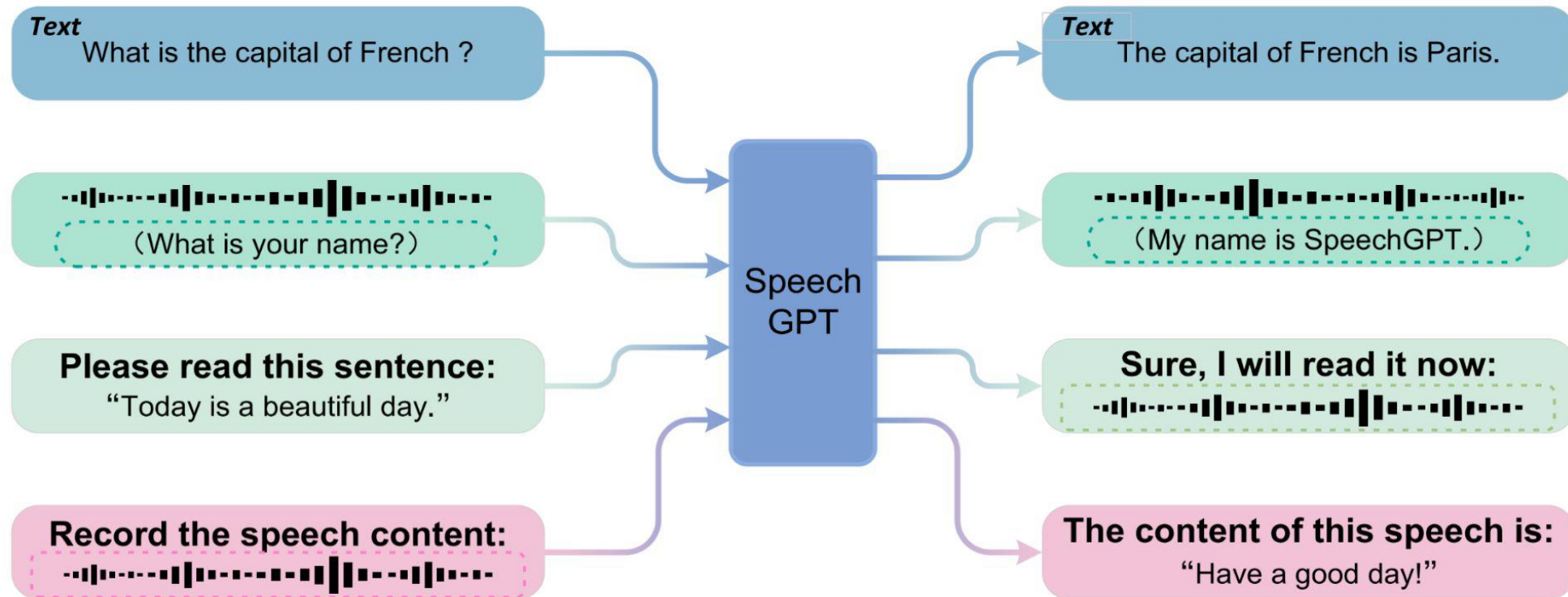
- Works for online decoding (streaming)
- Increment u when non-blank token is output
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- Works for online decoding (streaming)
- Increment u when non-blank token is output
- increment t when blank token is output



Capability



Website and Demo: <https://0nutation.github.io/SpeechGPT.github.io/>

Phonetics
Feature Extraction
Architectures

Word Error Rate

Definition
Significance Test

How to evaluate the word string output by a speech recognizer?

$$\text{Word Error Rate} = \frac{100 * (\text{Insertions} + \text{Substitutions} + \text{Deletions})}{\text{Total Word in Correct Transcript}}$$

REF: portable form of stores last night so

SYS: portable phone upstairs last night so

How to evaluate the word string output by a speech recognizer?

$$\text{Word Error Rate} = \frac{100 * (\text{Insertions} + \text{Substitutions} + \text{Deletions})}{\text{Total Word in Correct Transcript}}$$

REF:	portable	FORM	OF	STORES	last night so
SYS:	portable	*****	PHONE	UPSTAIRES	last night so
Eval:		I	S	S	

How to evaluate the word string output by a speech recognizer?

$$\text{Word Error Rate} = \frac{100 * (\text{Insertions} + \text{Substitutions} + \text{Deletions})}{\text{Total Word in Correct Transcript}}$$

REF:	portable	FORM	OF	STORES	last	night	so
SYS:	portable	*****	PHONE	UPSTAIRES	last	night	so
Eval:		I	S	S			

$$\text{WER} = 100 (1+2+0)/7 = 42.86\%$$

Word Error Rate (Whisper)

English Tasks	WER%
LibriSpeech audiobooks 960hour clean	1.4
LibriSpeech audiobooks 960hour other	2.6
Switchboard telephone conversations between strangers	5.8
CALLHOME telephone conversations between family	11
AMI meetings	11
Sociolinguistic interviews, CORAAL (AAE)	16.2
CHiME6 dinner parties with distant microphones	25.5
Sample tasks in other languages	WER%
Common Voice 15 Vietnamese	39.8
Common Voice 15 Swahili	51.2
FLEURS Bengali	50
Chinese (Mandarin) Tasks	CER%
AISHELL-1 Mandarin read speech corpus	3.9
HKUST Mandarin Chinese telephone conversations	18.5

- Errors might be function words (and, of, uh-huh,...)
- Should we be more concerned with meaning (“semantic error rate”)?
 - Good idea, but hard to agree on
 - Has been applied in dialogue systems, where desired semantic/task output is more clear

- Determining if two word error rates are significantly different
- Matched-Pair Sentence Segment Word Error test
 - Segments: Region bounded on both sides by words that both recognizers get correct

	I	II	III	IV
REF:	it was	the best of	times it was the worst	of times it was
SYS A:	ITS	the best of	times it IS the worst	of times OR it was
SYS B:	it was	the best	times it WON the TEST	of times it was

- Calculate $Z_i = N_A^i - N_B^i, i = 1, \dots, n$ with N_A^i, N_B^i number of errors for segment i by system A, B
 $Z = \{2, -1, -1, 1\}$
- Calculate $\mu_z = \sum_{i=1}^n Z_i / n$ $\sigma_z^2 = \frac{1}{n-1} \sum_{i=1}^n (Z_i - \mu_z)^2$ $w = \frac{\mu_z}{\sigma_z / \sqrt{n}}$
- Reject null hypothesis $H_0: \mu_z = 0$ if $2 * P(Z \geq |w|) \leq 0.05$
 Look up probabilities in the standard tables of the normal distribution

Phonetics

- Phones are speech sounds that can be represented with symbols
- Prominence, prosodic boundaries, and tune are aspects of prosody
- Phones have a specific spectral signature

Feature Extraction

- The audio signal is decomposed into Hamming windows
- For each window, frequency components are extracted using Fourier transform
- For each window, extract frequency band on the Mel scale

Architectures

- Encoder-decoder models can be used for speech recognition
- CTC can be used for online speech recognition
- To improve accuracy, RNN-transducer add a language model to CTC encoder

- Jurafsky Dan and Martin, James H. "Speech and Language Processing (3rd ed. draft)" Chapter 16
<https://web.stanford.edu/~jurafsky/slp3/>
- <https://haythamfayek.com/2016/04/21/speech-processing-for-machine-learning.html>
- <https://medium.com/analytics-vidhya/understanding-the-mel-spectrogram-fca2afa2ce53>
- <https://lorenlugosch.github.io/posts/2020/11/transducer/>
- <https://openai.com/research/whisper>